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CITATION NETWORKS IN THE LAW

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INTRODUCTION

As the number of court cases decided each day steadily grows lawyers find it increasingly difficult to find the necessary case law materials to solve a particular legal problem. Existing manual and automated retrieval systems are of some assistance, but none of them is entirely satisfactory. In this paper I propose some new approaches for solving the legal retrieval problem. The proposed retrieval strategy is a hybrid of concepts from the law and from the areas of information science and computer science.

Three independent hypotheses have been formulated and evaluated as a result of the research work which has been completed. These three hypotheses are

- (1) Law retrieval systems should be designed so that the cross citations that exist between cases are used to select relevant cases.
- (2) The factual context of a case can be determined by examining the first paragraph in the statement of facts portion of a case and the first paragraph in the opinion portion of a case.
- (3) The main issues in a law case are most likely to appear in sentences which contain the word "question" in combination with one of the words "whether," "hold," or "conclusion."

The evaluative studies for the three hypotheses indicate that each can be of use in designing a workable retrieval system. In each instance, however, further work will have to be done to ensure that the hypothesis is entirely valid. In addition to proposing and evaluating these hypotheses, I have indicated how these ideas can be used as a foundation for an interactive computer system. A sample problem utilizing this system is included.

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The Retrieval Pattern – Traditional

After ascertaining the facts surrounding the legal problem posed by a client, the lawyer must begin his search through the digests, indexes, and annotations to find relevant case law materials. Any useful cases found need to be examined to determine what subsequent cases use them as a basis for decision.¹ These cases are found by using the *Shepard's Citator*.²

Each of the cases that appear in the Shepard's lists is also read. Those irrelevant to the lawyer's search are discarded while those which are potentially relevant are themselves shepardized. This shepardization process continues until the lawyer either runs out of time, becomes satisfied with the set of cases he has found, or feels that the set of cases is not really relevant after all.

If the set of cases isolated in this manner is not helpful, the lawyer has two possible strategies to follow. He may choose either to continue searching in the indexes and in the annotations for a new set of cases or to look at some of the cases which are "cited in" the cases that have been isolated.

The process outlined above is so time-consuming that most lawyers do not follow it to its logical conclusion. That is to say, a lawyer will often do only a partial shepardization rather than follow all of the possible cross-citations to their logical end. Lawyers may also hesitate to do an exhaustive shepardization because it has not been shown that an exhaustive shepardization is a meaningful way to isolate a large enough set of highly relevant cases.

It is reasonable to hypothesize that the exhaustive shepardization process can be used to isolate a large set of cases that are relevant to a legal problem. The evaluative studies made on this hypothesis appear in the section below designated "The Retrieval Pattern – Proposed." I have developed some techniques which make it possible to accomplish the exhaustive shepardization quickly and effectively. These techniques are discussed in the section designated "An Operating System."

Retrieval Systems Presently Available

As indicated above, there are several other manual and automated systems available for assisting the lawyer in finding the necessary case law materials to solve his client's problems. Some of these systems are discussed briefly below.

Compilations of legal cases according to code numbers have been available since at least 1888. The most popular of these systems in use today is the West Key Number system.³ Any legal researcher will attest to the difficulty of using the *West General Digest*. The system is incredibly complex. There are, for example, at least four hundred (400) major classification headings, each with

1. The term *precedent* in this paper refers only to the fact that a case cites another case. We are unconcerned with notions of *stare decisis*. It does not matter that a judge has erroneously cited a case in reaching his decision. The problem of interpreting the law is outside the scope of this research. It is up to the lawyer to decide the wisdom of a particular decision.

2. SHEPARD'S UNITED STATES CITATIONS—CASES (5th ed. 1943); SHEPARD'S UNITED STATES CITATIONS—CASES (5th ed.), CASE EDITION SUPPLEMENT 1943-1964 (1964); SHEPARD'S UNITED STATES CITATIONS—CASES (5th ed.), CASE EDITION SUPPLEMENT 1964-1968 (1968); and SHEPARD'S UNITED STATES CITATIONS—CASES (5th ed., Parts I & II, April, 1969).

3. SEVENTH DECENNIAL DIGEST, AMERICAN DIGEST SYSTEM 1956-1966 (38 vols., 1967). And 9 WEST'S GENERAL DIGEST (4th series, 1969).

from sixteen (16) to two thousand (2000) subheadings. The *Decennial Digests* consist of at least thirty large volumes.

Another way of classifying cases is by the annotation method. Works such as *American Law Reports Annotated*⁴ are often used by lawyers. Annotations are generally not as complicated as the key word digests, which stuff the cases into so many finely divided pigeonholes. Moreover, the annotation is usually more illustrative than is the overly concise key word index. Nevertheless, annotations are highly structured and use key words for creating classifications.

There are four important drawbacks to systems based mainly on the use of key words and phrases: (1) these systems are static in their terminology and not adaptable to vocabulary changes; (2) these systems require that the user's thinking conform to the classifications formulated by the system designers; (3) these systems classify the law according to a rigid key word terminology without indicating the context in which the words appear; and (4) each of these systems is based on indexing and classifying that has been done by human indexers.

The indexers are not necessarily skilled lawyers. Moreover, they are faced with the almost insurmountable task of fitting particular cases into a set of inflexible and pre-arranged pigeonholes. Today, after more than a half century of anti-trust laws, many indexes still list anti-trust cases under the old heading of *monopoly*. Another example is Labor Law, which is classified under *Master-servant*. It would be too difficult (it seems) to create a new category.

Several automated systems have been implemented from time to time. They include the Law Research Service,⁵ the University of Pittsburgh Health Law Center System,⁶ and Project LITE (Legal Information Through Electronics). These pioneering systems have been of limited success because of the degree to which they rely on key words and phrases, and because they do not make full use of the interlocking structure of case citations.

The key word systems are based on the underlying concept that identifying key words is the best way to convey the informational content of the natural language text for retrieval purposes. A number of researchers have shown this approach to be inadequate.⁷ Basically, the problem is that the textual material is too rich and variegated to be classified in terms of a few key words. Moreover, the context in which a certain set of key words appears is very important, and the Law Research Service, for example, does not take this factor into account. The University of Pittsburgh System is somewhat more sophisticated in this area. This system provides the user with several reference tools that enable him to make a more intelligent search. But most of the searches made on the University of Pittsburgh System are on a case-by-case basis.

A proposal for a system that does not require a case-by-case examination of all the cases in the memory files is outlined below. This proposed system does not rely primarily on a pre-determined set of index terms. Rather, it makes limited use of the context in which a particular word appears.

4. AMERICAN LAW REPORTS ANNOTATED 3RD, CASES AND ANNOTATIONS (1965).

5. Law Research Service, N.Y., N.Y.

6. Materials obtained from Mr. J. Horthy, Health Law Center, University of Pittsburgh, Pittsburgh, Pennsylvania.

7. H.P. Edmundson and R.E. Wyllis, *Automatic Abstracting and Indexing—Survey and Recommendations*, in 4 COMMUNICATIONS OF THE ACM, No. 5 (1961).

Retrieval Pattern – Proposed

Presently available legal retrieval systems have failed to meet the needs of lawyers. The early systems are of use only in isolating a starting set of cases. The automated systems make it possible to select a starting case more quickly, but they do not always help the lawyer isolate a whole set of relevant cases.

The University of Pittsburgh System has been useful for retrieving cases, but it is too cumbersome to use when millions of cases are put into the memory files. The number of reported cases continues to mushroom each year.⁸

Traditional systems are deficient because either they try to use key words to classify complex legal issues or they try to use a case-by-case search of the entire memory file. It is often impossible to search the entire file in any reasonable amount of time.

None of the present automated systems employs the methodology that the lawyer actually uses when he does legal research. In other words, none of these systems attempts to make use of the cross-citation structure that is built into case law materials. The use of case citations is an inherent and fundamental characteristic of these materials. Judges have used citations throughout the entire legal history of the United States. Lawyers rely heavily on this cross-citation structure in putting together a cogent legal analysis.

As indicated above, lawyers use the cross-citation structure whenever they do legal research. But because of time considerations they do not perform what I call "exhaustive shepardization."

Exhaustive shepardization is the process of (a) selecting a case relevant to the legal problem which faces the lawyer and designating that case as the root case; (b) selecting all cases which cite the root case; (c) selecting all cases which cite the cases in (b), etc. until no more citation strings can be found; (d) selecting all cases which are cited *in* the cases collected in steps (a), (b), and (c); and (e) repeating steps numbered (b), (c), and (d) until no more cases are found.

Almost every case in the American legal system is tied to other cases in the system by the citational process. Very few cases exist which cite no other cases and which are not cited by other cases.

Instead of relying on the superimposed key word systems, the researcher should rely on the cross-citation structure because it provides a more rational linkage between relevant cases. The citational links exist because at some point in time a judge and a lawyer decided that a logical connection existed between certain cases.

Because a particular case might be cited for a number of different reasons, it is important to single out those citations that are relevant to the problem being researched. Thus the exhaustive shepardization process is really one of "selective" exhaustive shepardization. At each point in the shepardization, the researcher should determine if the cases being searched are relevant to the original problem. If they are not, they must be dropped from the group of collected cases.

If the lawyer wishes to select groups of cases that have the same factual context, the selective shepardization is particularly helpful. A lawyer whose

8. A rough estimate of the number of cases in the Northwestern University Law Library indicates that there are at least six million cases in print. See *A Proposal for Examining the Precedent Structure of the Law*, an unpublished paper written by the author of this paper as an independent studies project at Northwestern University.

client was injured in the course of surgery might want to find cases that apply the rule of *res ipsa loquitur* to this particular type of injury. Another example might be that of a client who wishes to produce a new giveaway show on television, but is unsure if the scheme is in violation of the anti-lottery laws. In this latter instance, the lawyer will want to find cases that define lotteries and give examples of illegal schemes. We shall return later to this second hypothetical situation.

I have developed some techniques which make it possible to obtain an exhaustive shepardization in a selective manner if the selection is based on the factual setting of the cases being shepardized. These techniques have been tested; the results appear below. At the present time I have not adequately solved the problem of distinguishing cases on the basis of statements of law. However, I have identified some important characteristics of the legal language which make it possible to identify those statements in a legal opinion which state the central legal issues in the case. These characteristics are also discussed below.

The proposed retrieval strategy is based on the notion that an exhaustive shepardization can be used to effectively retrieve a set of relevant case law materials if the lawyer begins with one relevant case. This is basically an hypothesis from the law.

A second aspect of the strategy is the notion that the shepardization process can be controlled in the area of factual discrimination so that irrelevant cases are filtered out. The filtering-out process is accomplished by using techniques from the information and computer sciences.

A third aspect of the strategy is the notion that some language clues effectively identify those sentences in an opinion which state key legal issues. This is essentially an information sciences hypothesis.

The three hypotheses outlined above are independent of one another. If one hypothesis is invalid it does not affect the utility of the other hypotheses. For example, assume that the exhaustive shepardization process does not really give the lawyer access to a satisfactory set of relevant case law materials. None the less, the proposed method for achieving selective shepardization on a factual basis is still a very useful tool.

Data Base

In order to test the validity of the three hypotheses, a data base had to be formed. An annotation was selected randomly from the *United States Supreme Court Digest Annotated*.⁹ The annotation selected is entitled "Lottery"; it appears in volume nine. The scope note of the annotation says it deals with the "civil and criminal aspects and incidents of lotteries, and of particular schemes, devices, and enterprises, constituting lotteries." The annotation cross-references two other annotations: "Post Office (mails)," and "Constitutional Law (constitutional guarantees)."

Also included in the annotation are thumbnail descriptions of sixteen cases. Since some of the cases are described more than once, there is a total of thirty-seven descriptive passages. These passages are divided into a number of categories (e.g., "Lottery grant or permit"; "What Constitutes"; and "Liability to ticket holder").

9. UNITED STATES SUPREME COURT DIGEST ANNOTATED, LAWYERS' EDITION (18 vols. 1969).

For each of the sixteen cases in the annotation a data file was made consisting of seven elements:

(1) *THUMBNAIL.*

The descriptive passage from the *United States Supreme Court Digest Annotated*.

(2) *HOLDING.*

A statement of the key issues of the case based upon a reading of the entire opinion.

(3) *ANCESTORS.*

A list of all cases cited in the opinion of the case.

(4) *DESCENDANTS.*

A list of every case which cites the case.

(5) *ANNOTATION.*

A list of all the cases that appear in the annotation that accompanies the case in the *Lawyers' Edition, United States Supreme Court Reports*.¹⁰

(6) *FIRST PARAGRAPH OF FACTS.*

The entire first paragraph of the case as reported in the *Lawyers' Edition*.*

(7) *FIRST PARAGRAPH.*

The entire first paragraph of the opinion portion of the case as reported in the *Lawyers' Edition*.

A similar file was created for every ancestor or descendant of the sixteen primary cases. Approximately one hundred and twenty Supreme Court cases were collected in a manner described in the next section. Cases from the lower federal courts and from the state courts were not analyzed, for two reasons: (1) the high degree of interconnectedness between just the Supreme Court cases indicated that the exhaustive shepardization approach is a meaningful one; (2) The omission of the state and lower federal court cases has no bearing on the validity of the other two hypotheses.

Evaluation of the Exhaustive Shepardization Process

In order to evaluate the extent to which the exhaustive shepardization process can be used to retrieve a large set of relevant case law materials, the following three evaluative tests were designed:

Test One

This test is designed to show that if the exhaustive shepardization process is applied to any case in the *Supreme Court Digest Annotated*, all the other cases which appear in its annotation will be retrieved.

Test Two

This test is designed to show that if the exhaustive shepardization process is

10. The text of all cases used in this study is from the *Lawyers' Edition, United States Supreme Court Reports*, hereinafter referred to as *Lawyers' Edition*.

* This element of the record file was empty if the paragraph said, "The facts of the case are reported in the opinion."

applied to annotated cases appearing in the *Lawyers' Edition*, all the relevant cases in the annotation will be retrieved.

Test Three

This test is designed to show that if the exhaustive shepardization process is applied to any case in the *Supreme Court Digest Annotated*, all the relevant cases that appear in related annotations will be retrieved.

The rationale behind each of these three tests is that the annotations in the *Supreme Court Digest Annotated* and in the *Lawyers' Edition* are in essence nothing more than a compendium of cases that cite one another. These annotations have been compiled by legal researchers because they show which cases are logically connected to one another. But if the exhaustive shepardization process can be used to isolate and identify the same set of cases, then that process is a reliable method for collecting a large set of relevant case law materials. I have shown that the cross-reference network conveys at least as much information as the annotations do.

TEST ONE

The exhaustive shepardization process can be used to retrieve the same set of cases that have been collected by the annotators in the *Supreme Court Digest Annotated*.

Each of the sixteen cases in the "Lottery" annotation was shepardized exhaustively. The results are shown in Figure 1, a schematic diagram of the links that exist between the original sixteen cases.¹¹

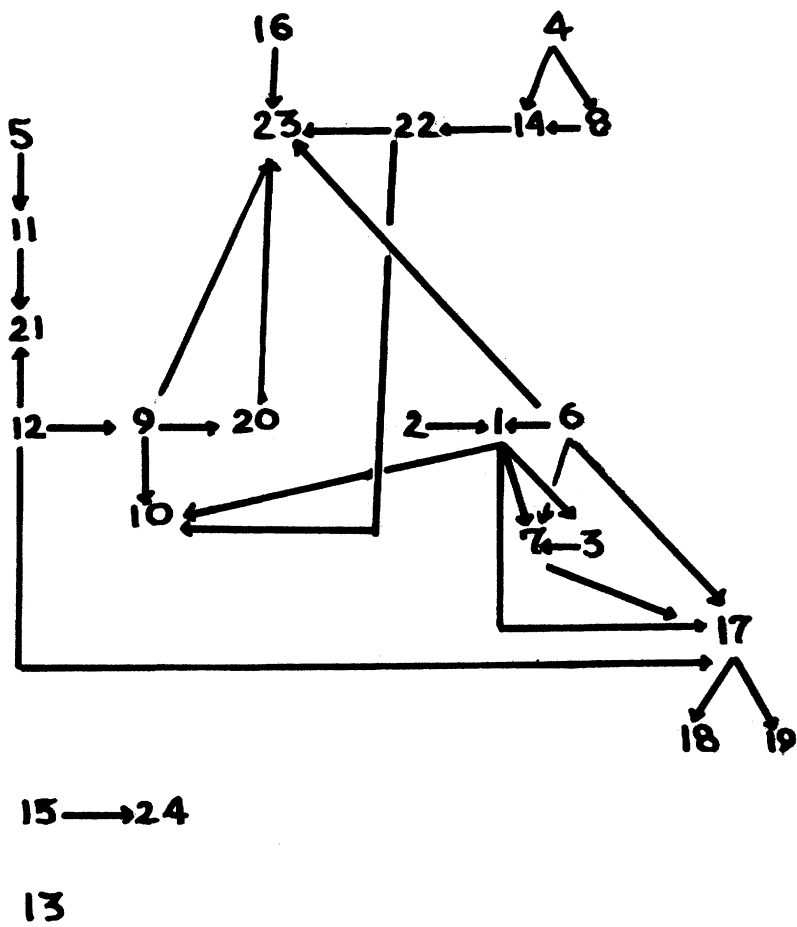
Cases that are from the "Lottery" annotation are numbered one (1) through sixteen (16) while other Supreme Court cases that serve as linkages are numbered seventeen (17) through twenty-four (24).

Only two of the original cases from the annotation are not linked to the rest (i.e., cases 13 and 15). The other fourteen are somehow interlinked to one another. There are several reasons why these two cases do not interconnect. Case number 13 is a very early case that is not cited by any subsequent Supreme Court cases, and it does not cite any cases. Case number 15 is also a very early case, and it does not cite any Supreme Court cases. The one case that cites case 15 is case number 24. Case 24, however, is a companion case to case 15; and it is not connected to any other cases.

11. The numbers in the diagram refer to cases in the following way: [1] *Stone v. Mississippi*, 101 U.S. 814 (1879); [2] *Boyd v. Alabama*, 94 U.S. 645, 24 L. Ed. 302 (1876); [3] *New Orleans v. Houston*, 119 U.S. 265, 30 L. Ed. 411 (1886); [4] *France v. United States*, 164 U.S. 676, 41 L. Ed. 595 (1896); [5] *Clark v. Washington*, 25 U.S. 40, 6 L. Ed. 544 (1827); [6] *Phalen v. Virginia*, 49 U.S. 163, 12 L. Ed. 1030 (1850); [7] *Douglas v. Kentucky*, 168 U.S. 488, 42 L. Ed. 553 (1897); [8] *Fabrizio v. United States*, 385 U.S. 263, 17 L. Ed. 2d 351 (1966); [9] *United States v. Horner*, 147 U.S. 449, 37 L. Ed. 237 (1892); [10] *FCC v. ABC*, 347 U.S. 284, 98 L. Ed. 699 (1954); [11] *Shankland v. Washington*, 30 U.S. 390, 8 L. Ed. 166 (1831); [12] *Cohens v. Virginia*, 19 U.S. 264, 5 L. Ed. 257 (1821); [13] *Catts v. Phalen*, 43 U.S. 376, 11 L. Ed. 306 (1844); [14] *Francis v. United States*, 188 U.S. 375, 47 L. Ed. 508 (1903); [15] *Brent v. Davis*, 23 U.S. 395, 6 L. Ed. 350 (1825); *in re Gregory*, 219 U.S. 210, 55 L. Ed. 184 (1910); [17] *Lottery Case*, 188 U.S. 321 (1903); [18] *United States v. Kahriger*, 345 U.S. 22 (1952); [19] *United States v. Badders*, 240 U.S. 391 (1915); [20] *Public Clearing House v. Coyne*, 194 U.S. 497 (1904); [21] *Suydam v. Williamson*, 61 U.S. 427 (1857); [22] *United States v. Halseth*, 342 U.S. 277 (1951); [23] *ABC v. United States*, 110 F. Supp. 374 (1953); [24] *Moses v. Washington*, 23 U.S. 406 (1825).

Two further points must be made about Figure 1. First, case number 23 is not a Supreme Court case but the lower court opinion of case number 16. Second, case number 21 is a questionable link in the graph because it does not deal with problems directly within the scope of the "Lottery" annotation. Thus it is arguable that only twelve of the sixteen cases are linked to one another. However, it must be pointed out that if the analysis is extended to encompass lower court opinions, many of the missing linkages may be uncovered.

FIGURE 1



The notation 1-2 means that case 1 cites case 2 as precedent. All relevant interconnections are shown. The graph is really a tree structure of finite size, because the legal system contains only a finite number of cases. The structure grows in one direction and is without loops because cases are ordered chronologically so that no case can cite cases decided after it.

TEST TWO

The exhaustive shepardization process can be used to retrieve the same set of cases that have been collected by the annotators in the *Lawyers' Edition Annotated*.

Ten of the sixteen cases from the "Lottery" annotation are reported in the *Lawyer's Edition Annotated* accompanied by a special annotation for that case. The exhaustive shepardization process was applied to each of the principal ten cases. In every instance the exhaustive shepardization process uncovered every case that was listed in the special annotation. Moreover, in seven instances even more cases were found by the exhaustive shepardization. These results appear in tabular form in Figure 2.

FIGURE 2

CASE CODE	NUMBER OF SUPREME COURT CASES IN ANNOTATION	NUMBER OF CASES FOUND BY BOTH SHEPARDIZATION AND ANNOTATORS	NUMBER OF CASES FOUND ONLY BY SHEPARDIZATION
1	13	13	9
2	3	3	10
3	3	3	3
4	1	1	0
5	3	3	2
6	5	5	1
7	6	6	14
13	1	1	0
11	6	6	1
15	1	1	0

TEST THREE

The exhaustive shepardization process can be used to retrieve the same set of cases that have been collected by annotators and categorized under related annotations in the *Supreme Court Digest Annotated*.

Five annotations in the *Supreme Court Digest Annotated* are related to the "Lottery" annotation. The cases from the "Lottery" annotation were exhaustively shepardized and the isolated cases compared to the cases listed in the five related annotations. The results appear in tabular form in Figure 3.

The results of test three indicate that the cases in the related annotations are linked to the cases in the "Lottery" annotation in most instances. Thus by starting with one of the cases in the related annotation one could find as many as fourteen of the cases included in the "Lottery" annotation without even knowing it existed.

The results of all three tests support the hypothesis that the exhaustive shepardization process is a good method for isolating a set of case law materials that are relevant to a particular legal problem. The process is more reliable than

the annotations that have been compiled by legal researchers. In several instances the annotations fail to give as much information as the cross-citation structure does.¹²

I have shown that the exhaustive shepardization process is a powerful and reasonable method for finding relevant case law materials. The process should be valid even when it is applied to annotations from other areas of the law. The materials examined herein have no specific characteristics which would make the results achieved unique. Hence, the fact that only one area of the law was used to test this hypothesis does not necessarily mean it is a weak one.

Further tests could be run, however. For example, an attempt might be made to see whether the exhaustive shepardization process can be used to identify the same set of cases as were found by an attorney during his research for an actual court case.

A law case may stand for a number of distinct legal and factual premises. An attorney who is using the exhaustive shepardization process needs a weeding out process so that he will not shepardize irrelevant cases. In the next section I will test a method for weeding out cases that have a factual basis different from that of the main case being shepardized.

FIGURE 3

Annotation	Number of Cases in Annotation	Number of Cases Found By Shepardization	Number of Cases Not Found
Post Office	6	5	1
Constitutional Law	2	1	1
Constitutional Law (Lotteries)	3	3	0
Constitutional Law (Lotteries—Impairment of Contracts)	3	3	0
Constitutional Law (Lotteries—Impairment of Contracts—What Lotteries Protected)	2	2	0
Totals	16	14	2

Determinations of the Factual Context of a Legal Opinion

For many years information scientists have endeavored to find ways of creating abstracts of natural language textual materials. An abstract is fundamentally designed to provide “a description of, or restatement of the essential content of

12. The most glaring example appears in the SUPREME COURT DIGEST ANNOTATED in its “Lottery” annotation. The annotators neglected to include the famous Lottery Case. This case was found, however, by the exhaustive shepardization.

a document.”¹³ The lawyer who writes down the factual context of a particular case is in essence creating a partial abstract of the case. This type of abstract is usually called a “functional” abstract.¹⁴ More specifically, it performs an “indicative function” because it “indicates” what the factual context of the case is.¹⁵

I contend that the factual setting of a legal opinion can be determined by using (1) the first paragraph in the statement of facts portion of a case and (2) the first paragraph in the opinion portion of a case.* In essence this is a technique for automatically creating an abstract. It must be emphasized that these two paragraphs can be used to determine the factual setting of a case only for purposes of deciding whether it is factually consistent with another case which it cites as precedent.

The selection of these two paragraphs is the result of extensive readings and re-readings of the case law materials in the data set. This choice takes advantage of the inherent characteristics, or “textual criteria,” of the case law materials.¹⁶

The lawyer and the judge are so trained to think that the first question they always ask is “What are the facts?” It is reasonable to argue that the first thing a judge will write, when handing down a legal opinion, is a statement of the facts in the case. It is reasonable that the inherent logic of the law (in requiring that the facts be given primary consideration) will dictate a special organization of judicial opinions. Judges should state the essential facts at the beginning of each opinion. This breeds a standard writing format, a “textual criteria,” for legal opinions.

This approach is similar to the one Baxendale used in her attempts to create abstracts automatically.¹⁷ She arbitrarily selected the first and last sentences in each paragraph of a document as components of its abstract. Baxendale did not state any reasons to support her choice of rules. (One could argue that good writers tend to state the most important ideas in the topic and closing sentences of each paragraph, but this proposition is unsupported by empirical evidence.) The selection herein is grounded in the basic logic of the law itself. It involves more than a standard of good writing: it involves the recognition of a rule of sound legal analysis.

For each of the one hundred and twenty cases that made up the data base the following test was performed:

By a reading of the first paragraph in the statement of facts portion of the case and a reading of the first paragraph in the opinion portion of the case, is it possible to discern the factual context of the case for purposes of deciding whether it deals with material covered by the scope note in the “Lottery” annotation of the *Supreme Court Digest Annotated*?

The results of this test appear in Figure 4. The first portion of the table (Case Code 1) is read as follows: (a) By a reading of the Facts paragraph alone it is possible to identify the factual setting of the cases in twenty-three instances; it is

13. R.E. Wyllis, *Extracting and Abstracting by Computer*, in AUTOMATED LANGUAGE PROCESSING, THE STATE OF THE ART 127-179 (H. Borko ed. 1967).

14. *Supra* note 7.

15. S.M. Marx, Automatic Abstracting (Unpublished paper, Northwestern University Library at Vogelback Computing Center, May, 1968).

* In Supreme Court opinions there are usually two sections that are designated as “Statement of facts” and “Opinion.”

16. *Supra* note 7.

17. *Supra* notes 7 and 13.

impossible in four instances; and it is undecidable in one instance. Two of the cases had no Facts paragraph. (b) By a reading of the Opinion paragraph it is possible to identify the factual setting in twenty instances; it is impossible in five instances; and it is undecidable in three instances. All cases had Opinion paragraphs. (c) By a reading of both the Facts and Opinion paragraphs in conjunction with one another it was possible to identify the factual setting of the case in all twenty-eight instances.*

These results indicate that the two paragraphs when read together are an effective tool for determining the factual setting of a legal opinion. Notice that the results could have been even better if more judges were to include a paragraph outlining the essential facts of the case.

It should be pointed out that the cases used in this analysis cover a time span of over one hundred and forty years. Over that long period of time this fundamental rule of legal writing has remained fairly static. There is little reason to assume it will change in the future.

FIGURE 4

Case Code	Cases Which Cite The Case Being Shepardized	Possible	Impossible	Undecidable	No Facts Paragraph
1	Facts Paragraph Only	23	4	1	2
	Opinion Paragraph Only	20	5	3	0
	Facts & Opinion Combined	28	0	0	2
3	Facts Paragraph Only	4	3	1	1
	Opinion Paragraph Only	6	2	0	0
	Facts & Opinion Combined	7	1	0	1
7	Facts Paragraph Only	15	6	1	4
	Opinion Paragraph Only	11	8	3	0
	Facts & Opinion Combined	21	1	0	4
10	Facts Paragraph Only	0	5	0	5
	Opinion Paragraph Only	4	0	1	0
	Facts & Opinion Combined	4	1	0	5
11	Facts Paragraph Only	3	4	0	3
	Opinion Paragraph Only	4	3	0	0
	Facts And Opinion	5	2	0	3

* The results of the combined readings are better than the results achieved by adding the separate readings together because the two paragraphs together provide “context” for one another.

An Operating System

I have outlined above the hypothesis that a legal information retrieval system should use the structure of the citation network. I have also outlined a technique for discriminating between cases on a factual basis. I should now like to outline a practical operating system which incorporates these concepts.

This information processing system is designed to operate in an interactive environment. Thus the lawyer and the computer will converse with one another through a remote console.¹⁸ In order to grasp more clearly how this system would operate, consider the following facts:

The client approaches the lawyer and outlines his plans for creating a new television giveaway show. The client wishes to know if the scheme is legal.

The lawyer will basically be interested in finding out which schemes fall within the prohibitory language of the anti-lottery laws. The lawyer may consult the *Supreme Court Digest Annotated* under "Lottery." Since all of the cases seem to be equally relevant he may just arbitrarily select *Stone v. Mississippi* (the first case) since it appears under the classification "General."*

The lawyer will have to read this case in order to get an idea of what it is about, and if it appears to be useful at all, he will be ready to use the system.

At the remote console the lawyer will be asked to signify the name of the starting case he has chosen as well as a list of words that describe the factual context of the case. These words will be placed on a list called the "yes" list, and the starting case will be designated as the "root" case.

The words that are on the yes list are "representative" of the facts in the root case and are selected by use of the following two rules:

Rule One: Select words from the scope note of the annotation, and from its sub-division headings.

Rule Two: Select words from the special pair of paragraphs appearing at the beginnings of the fact and opinion portions of the case.

The words selected for the yes list are chosen according to rules that have a foundation in logic. Namely, they are words that indicate the factual context of a case because they are found in the portion of the opinion which describes the factual content of the case or because they are found in the portion of the annotation that describes the reason that the cases are in the annotation.

The earliest information scientists, like Luhn, attempted to use selected words and phrases in the creation of automatic abstracts.¹⁹ But this type of selection has not proved useful because the words were chosen without recording the context in which they appeared. As indicated above, the words chosen for the yes list do have a special context.

The system will selectively shepardize the root case by comparing words on the yes list with words in the special paragraph pair in each case which cites the

18. An interactive system is one which allows the user to enter into a dialogue with the computer system. The user thus has the ability to intercede and alter the search strategy if he feels it is necessary. This type of system is desirable when complicated information like the law is being retrieved. It becomes possible for the lawyer to inject qualitative decisions into the search strategy.

* Actually this choice is not as arbitrary as it might seem at first glance. The same results would be reached (as will be shown) if the lawyer had just decided to begin with the famous *Lottery Case*.

19. *Supra* notes 7 and 13.

root case. If there are no matches (words in common) the words that appear in these cases will be placed on a "no" list. Thus the words on the yes list are words that one wants to find, while words on the no list are words that indicate an inconsistency with the root case. The rules that dictate how these lists are to be used appear in schematic form in Figure 5. The headings "Match on Yes List" and "Match on No List" mean that words in the special paragraph pair were matched against the list elements. The heading "Action" indicates whether a case is to be "accepted" as relevant or "rejected." The heading "Automatic or Manual" means that the rule calls for an action to be taken automatically by the computer system or that action is to be taken only on operator command.

In the sample problem, only four of the twenty-eight cases which cite the root case were selected using this process. These are the same four relevant cases as were identified during a reading of the entire opinions.

The system must next shepardize the new cases which have been isolated. In order to do this the yes and no lists must be updated. A check is made to see that the two lists do not have any words in common. The yes and no lists are then ready to be used for the next phase of the exhaustive shepardization process. This process continues until no further cases are found.

The rules outlined above were tested using the collected data base and the hypothetical problem outlined above. I acted as the computer and tried fairly to apply the rules that have been set out. In the beginning stages of the search process many of the decisions about the disposition of each case had to be made "manually." But as the search progressed the number of elements on the no list increased, and it became a useful tool for automatically eliminating cases. Gradually the system was able to assume almost the entire burden for deciding which cases were to be accepted. For example, in the first portion of the shepardization 50 percent of the decisions were automatic. In the second portion 60 percent were automatic. In the third portion 77 percent were automatic.

FIGURE 5

Rule	Match on Yes List	Match on No List	Action	Auto. or Manual
One	No	No	Accept/Reject	Manual
Two	No	Yes	Reject	Automatic
Three	Yes	No	Accept/Reject	Manual
Four	Yes	Yes	Accept/Reject	Automatic*

If a case has been automatically accepted or rejected the yes and no lists are not altered in any way. One of the greatest sources of error is to blindly add words to the lists.

If a case is manually handled, the user may alter the yes and no lists.

* Accept if yes matches are more numerous than no matches. Reject if no matches are more numerous than yes matches.

In only one portion of the search did the number of manual decisions increase again. That branch involved a series of cases that dealt with television giveaway shows. This line of cases turned out to be the desired goal of the search. These cases were not accepted automatically simply because they represent the modern line of lottery cases. The older lottery cases were decided before the words *radio* and *television* even existed; hence they were not originally on the yes list.

Using this operating system I was able to identify the same set of lottery cases as were identified by reading the entire case opinions. Thus the yes and no lists appear to be useful tools in performing a selective shepardization. The process requires a relatively low amount of human intervention while remaining fairly reliable.

Before this type of system could be implemented it would have to be expanded considerably. It should include a mechanism for parsing sentences which contain the words used on the yes and no lists.²⁰ It might also be expanded to contain a dictionary and thesaurus to give the user a clearer idea of the meanings of specific terms.²¹ In other words, features would have to be added so that the system would not be based so heavily on key words.

Sentences Containing Major Legal Issues

One rule developed by information scientists is that there are certain "syntactic" clues that indicate which words or phrases contain important information.²² These clues can be used to help identify the sentences in case law material that state major legal issues. There appear to be combinations of the words "question," "whether," "hold," "held," and "conclusion" that signal the reader that the judge is stating the key legal issues in the case. This hypothesis was tested as outlined below.

The entire record file for each of the one hundred and twenty cases was read, and each sentence containing the words "question" or "whether" was extracted. Ninety-five sentences were found. Each sentence was then read (out of its context) to determine if it stated (1) the major issue of the case it came from, (2) a minor issue of the case it came from, or (3) no issue at all. Some sample sentences from each of the three categories appear below:

Major Issues

The primary QUESTION . . . is WHETHER an officer appointed for a definite time during good behavior had any vested interest of contract right of which Congress could not deprive him.

The QUESTION for decision is WHETHER a tax laid by California on the receipt by defendant of the reinsurance premiums during 1930-31, infringes the due process clause of the 14th amendment.

The federal QUESTION presented by this case is WHETHER state statutes subjecting the property of a railroad corporation to taxation impair the obligation of the contract of the county charter.

20. Bertrand Raphael, *SIR—Semantic Information Retrieval*, in *SEMANTIC INFORMATION PROCESSING* (Marvin Minsky ed. 1969). This dissertation indicates some new and exciting ways for analyzing textual material in sentence form on the computer.

21. Morris Rubinoff's systems for retrieval of toxicological information at the University of Pennsylvania illustrate that dictionaries and thesauri and other user-oriented tools make possible extremely powerful retrieval systems.

22. *Supra* notes 7 and 13.

Minor Issues

The single OUESTION therefore, presented upon the record is WHETHER, the statute of Iowa set out in the plea, constitutes a defense to the action.

(Note: this cannot be a major issue because we have no idea what the offense is.)

Some cases turned on the QUESTION as to the city's want of legislative power to make the grant in the first instance.

No Issue

In our opinion these QUESTIONS should be answered in the negative.

The cases raise the same federal QUESTIONS.

In Figure 6 the results of the test appear in tabular form. These results indicate that the appearance of the word "question" in combination with certain special words signals the reader that the sentence contains a major legal issue. This type of analysis of legal writing is useful for creating new ways of automatically abstracting case law materials.

FIGURE 6

Sentence Type	Sentence States Major Issue	Sentence States Minor Issue	Sentence States No Issue
Sentence contains QUESTION and either HOLD, HELD, or CONCLUSION	62%	35%	3%
Sentence contains only the word QUESTION	3%	25%	72%
Sentence contains only the word WHETHER	5%	25%	60%

Conclusions

A legal information retrieval system must make intelligent use of the characteristics of the documents it retrieves. This paper is aimed at pointing out important clues available to information scientists who wish to develop a workable legal retrieval system.

The citation network of the law can be used to identify a set of cases highly relevant to the solution of a particular problem. The use of this structure has been ignored by most designers of automatic retrieval systems. Most of the systems rely on a tedious and time-consuming case-by-case search. Many of the systems use key words and other pre-coding schemes that ignore the rich context in which the words appear. A system which uses the cross-citation structure of the law can avoid many of these pitfalls.

A technique has been developed for automatically isolating the factual context of a case. This technique involves creating a partial abstract of the case, a special abstract based on rules of good legal logic and good legal writing. A proposal has been made for implementing the above hypotheses into a workable interactive computer system. A sample problem using that system has been simulated and the results evaluated. Finally, an attempt has been made to delineate those sentences in a legal opinion which state the basic legal issues in the case.